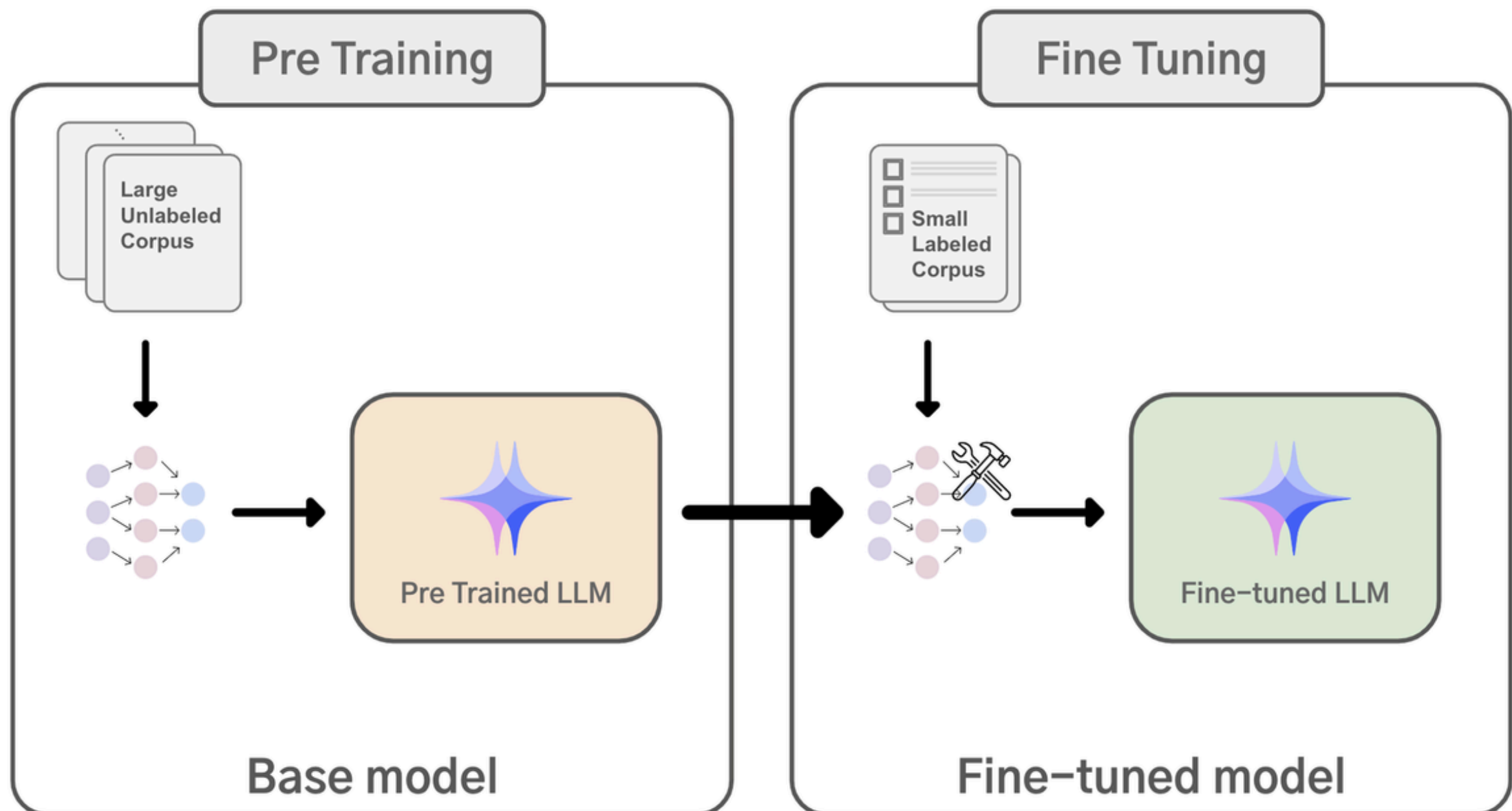


Finetuning Dictionary

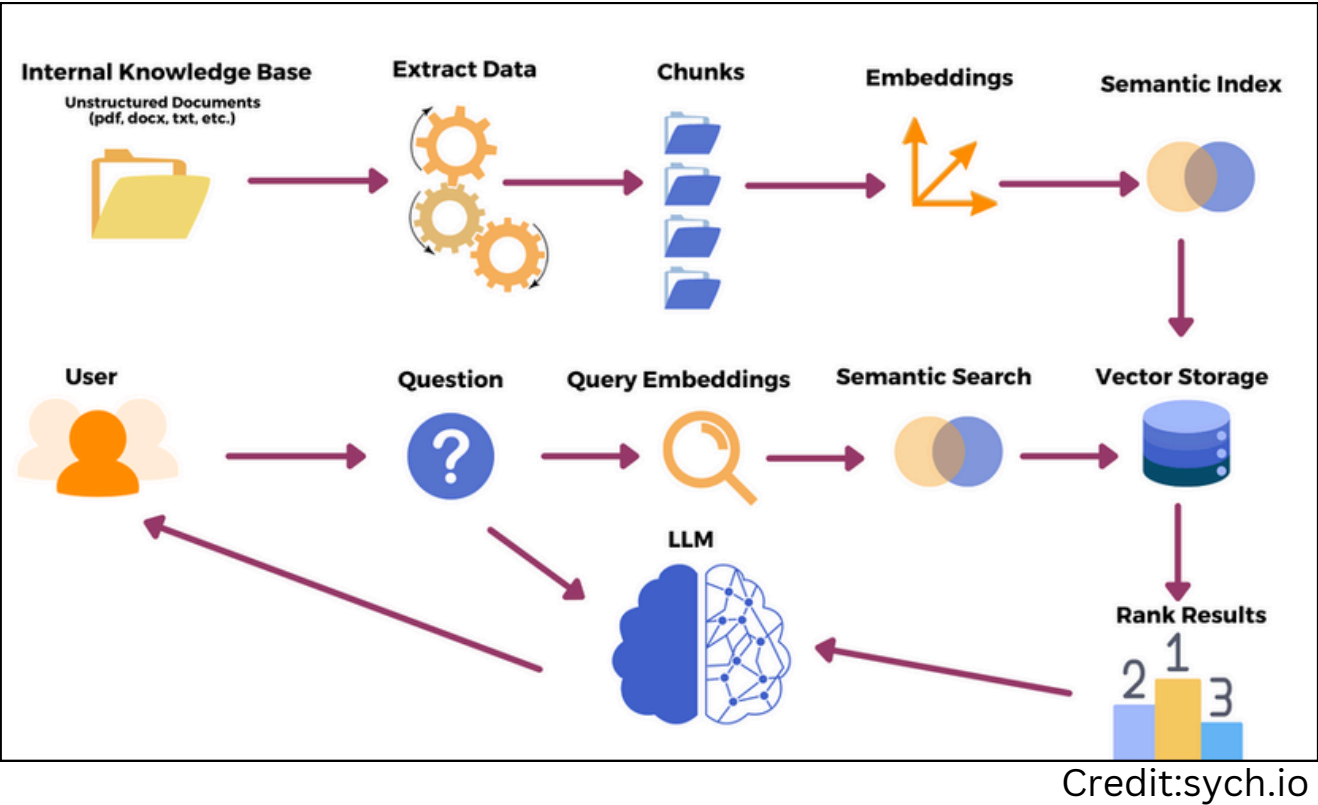
Your A-to-Z Guide to Mastering LLM Finetuning



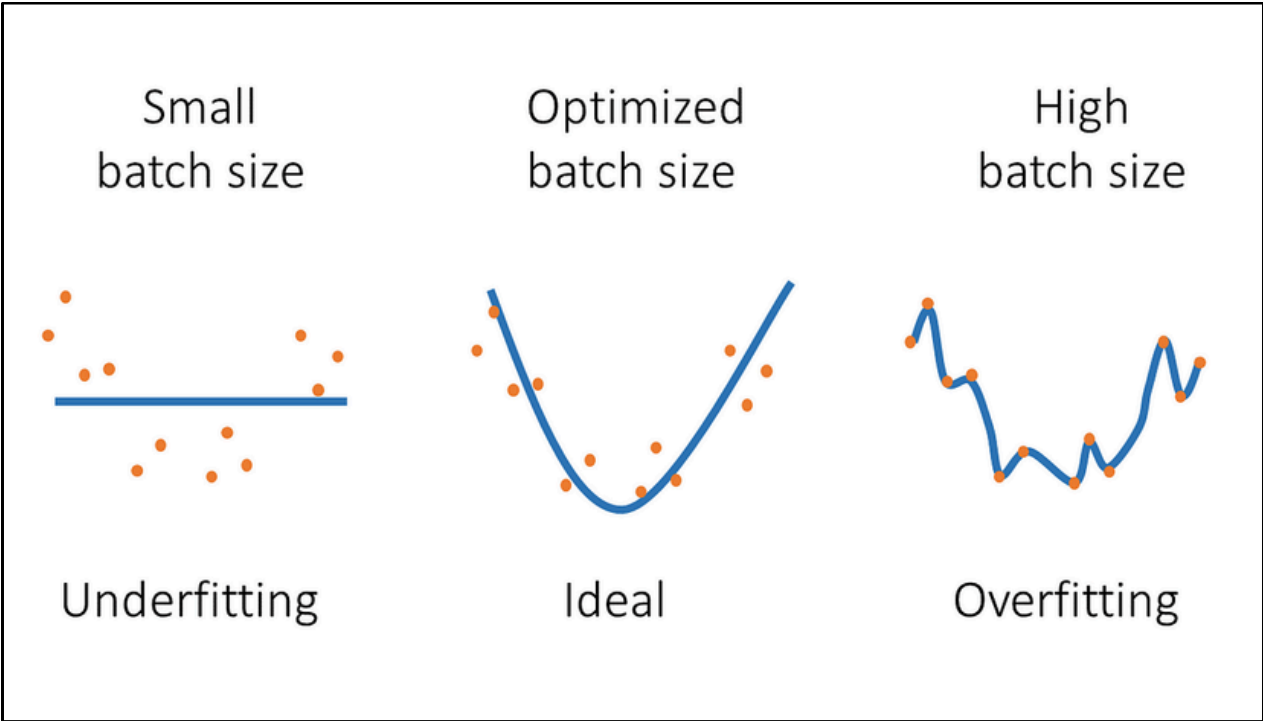
A - Augmentation

Expanding model abilities with external data or techniques.

- Ensures better outputs by adding domain knowledge.
- Enhances reliability in fine-tuned LLMs.
- Common in adaptive learning systems.



B - Batch Size



Source

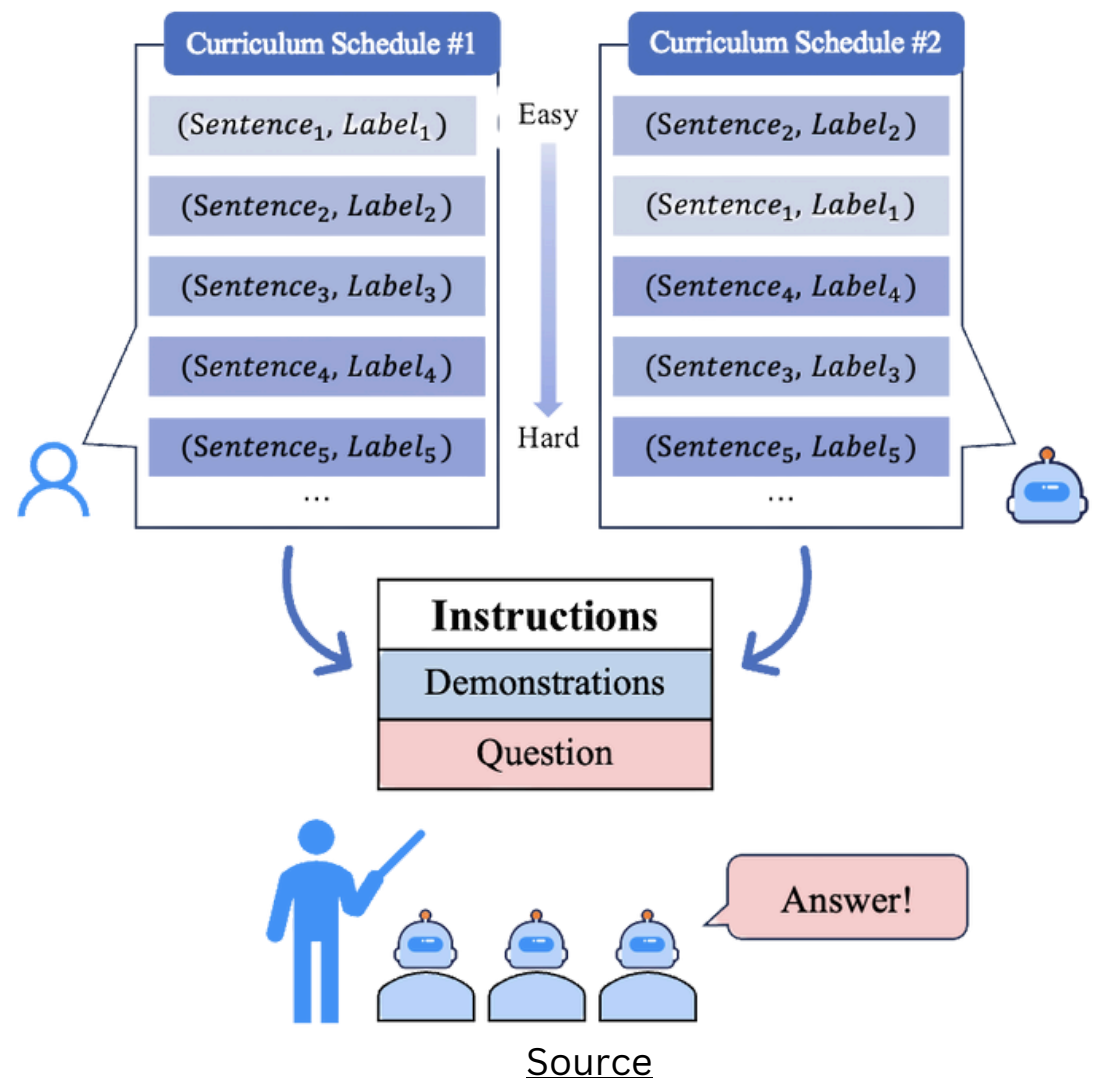
The number of samples processed before updating model weights.

- Affects learning stability and speed.
- Balancing small and large batch sizes is key.
- Important for tuning computational efficiency.

C - Curriculum Learning

Training models step-by-step with increasing complexity.

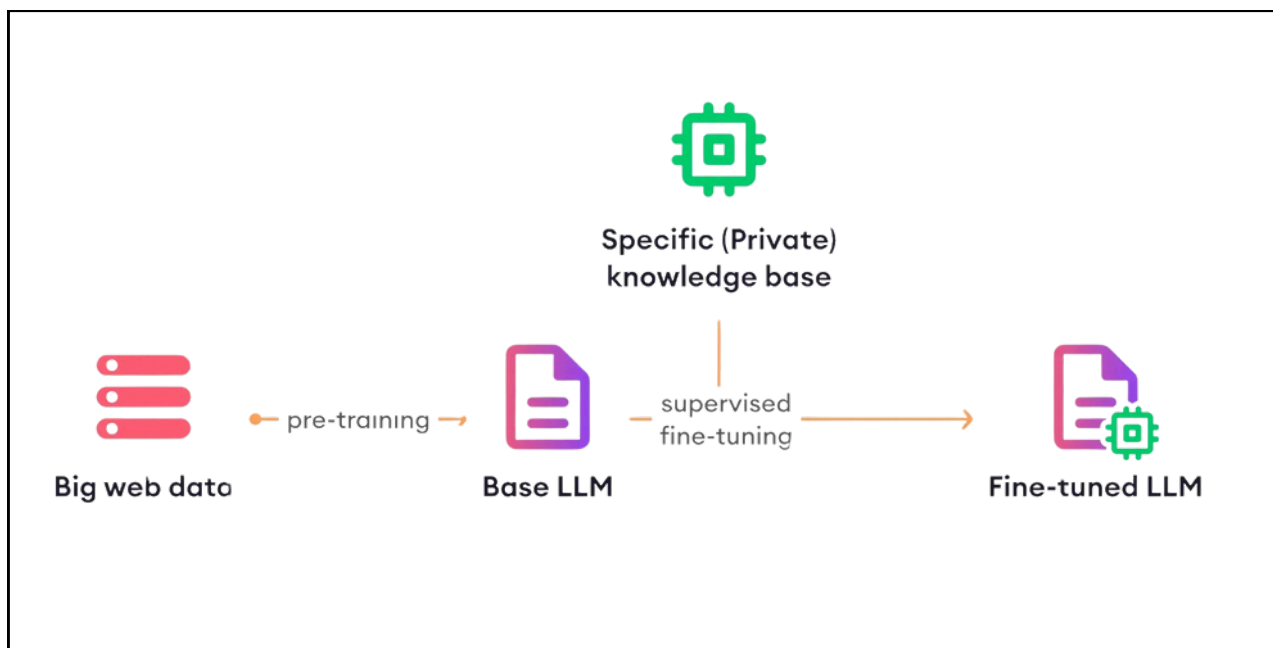
- Speeds up convergence.
- Helps the model generalize better to tasks.
- Inspired by human learning systems.



D - Domain-Specific Tuning

Adapting LLMs to excel in specialized fields.

- Essential for medical, legal, or finance sectors.
- Improves performance in niche applications.
- Requires high-quality labeled datasets.



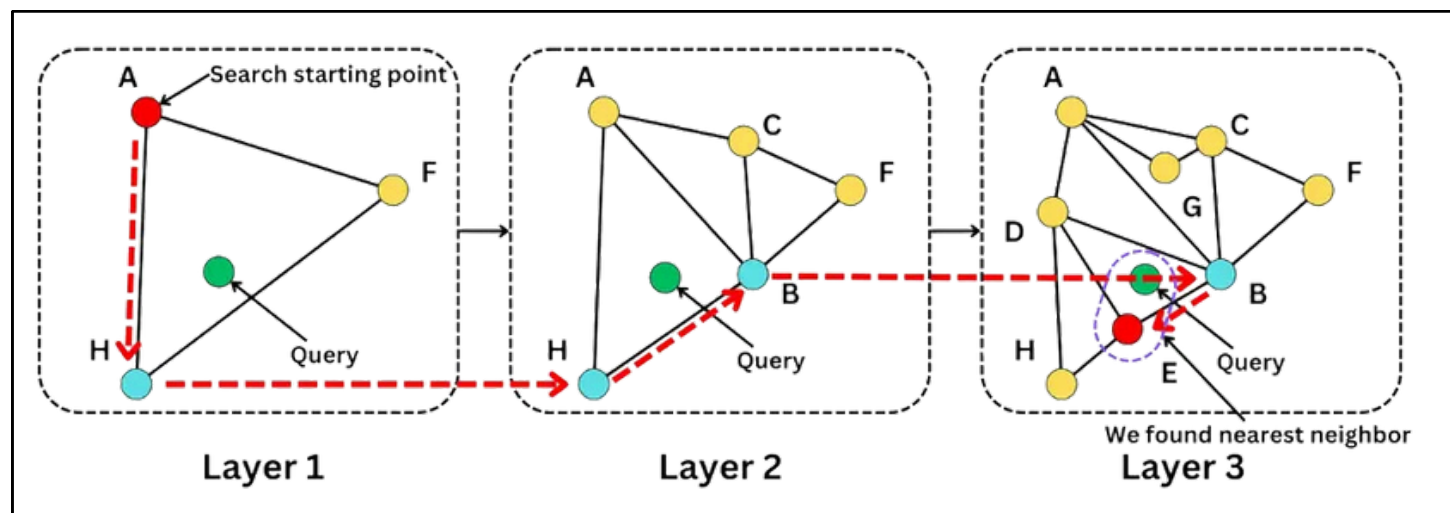
Source

*self promotion ahead

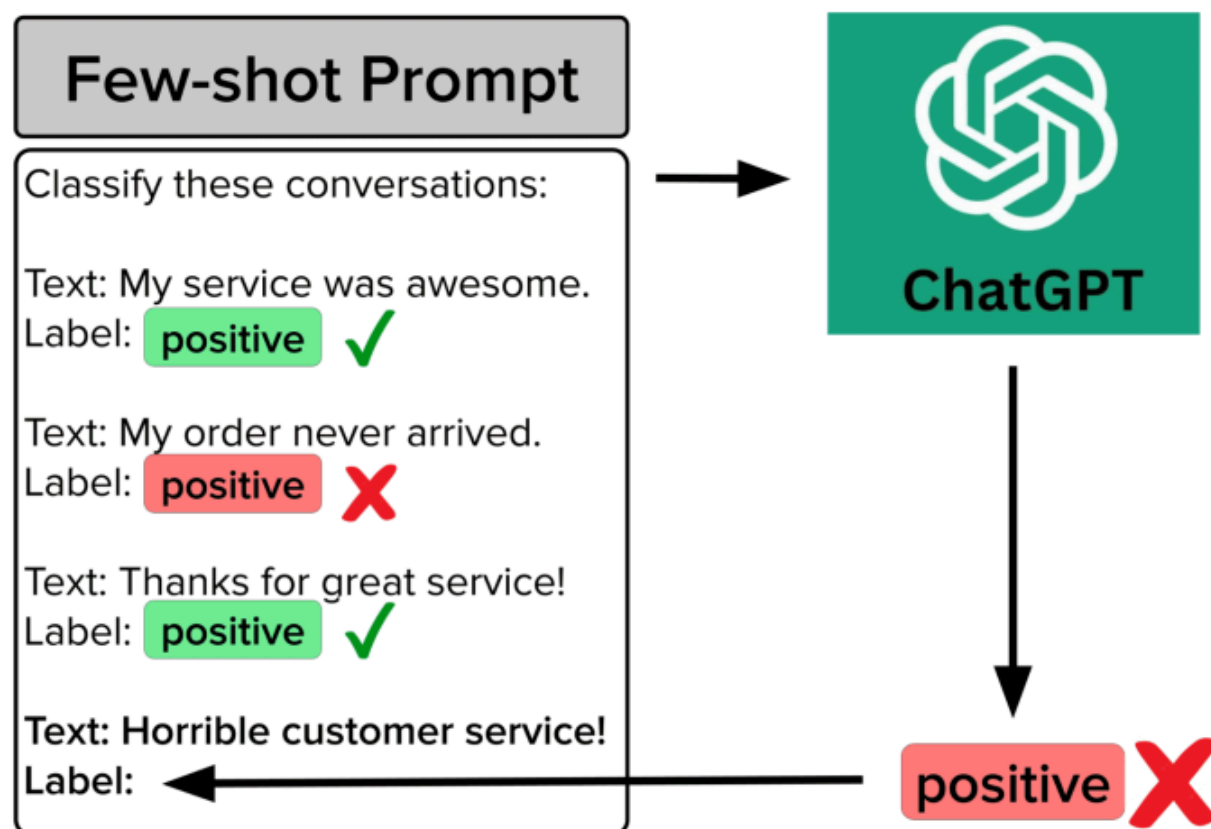
E - Embeddings

Converting text into numerical vectors for analysis.

- Key for semantic search & dense retrieval.
- Used extensively in fine-tuning pipelines.
- Powers tasks like recommendations and clustering.



F - Few-Shot Learning



Fine-tuning with minimal labeled data.

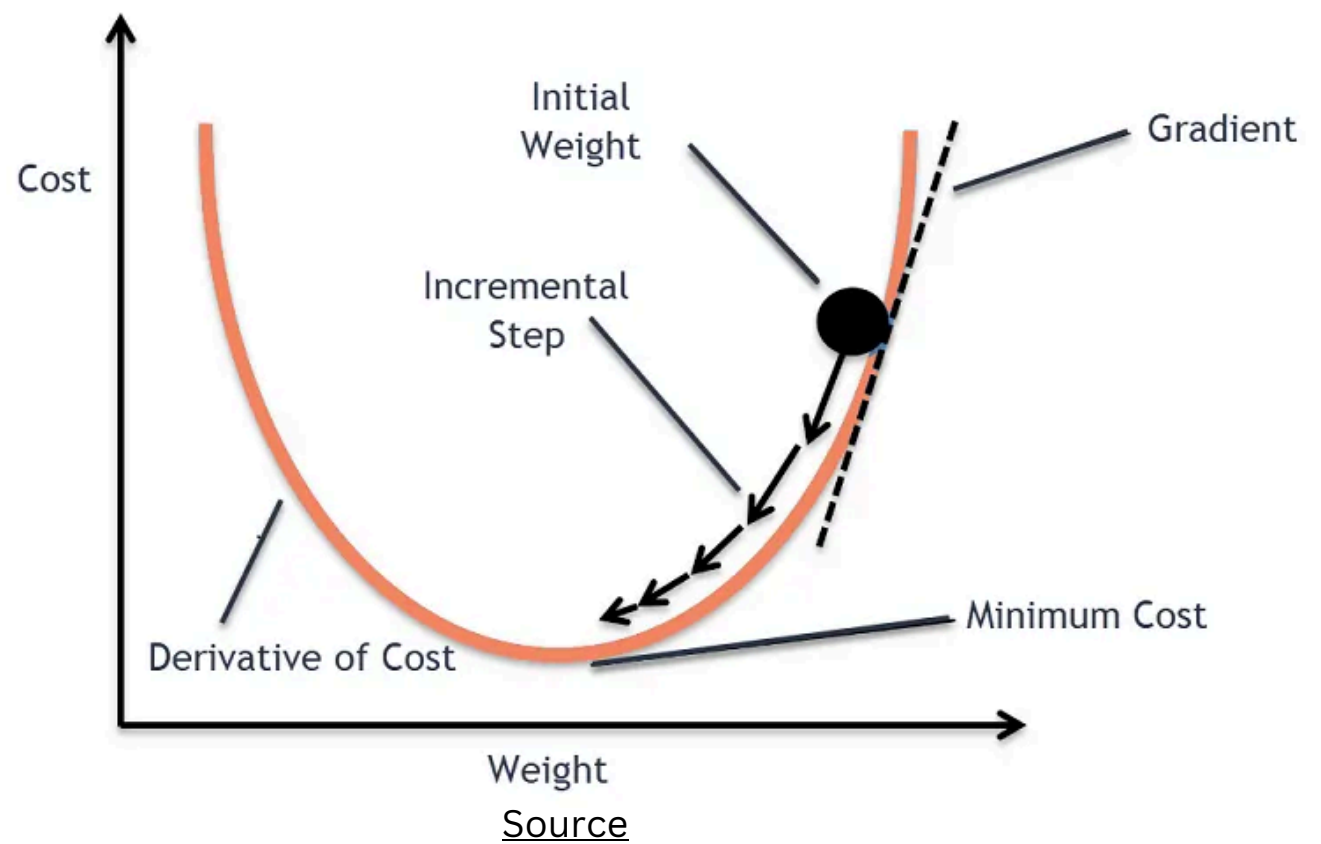
- Enables rapid adaptation to new tasks.
- Helps in low-data scenarios.
- Balances pre-training and specialized knowledge.

[Source](#)

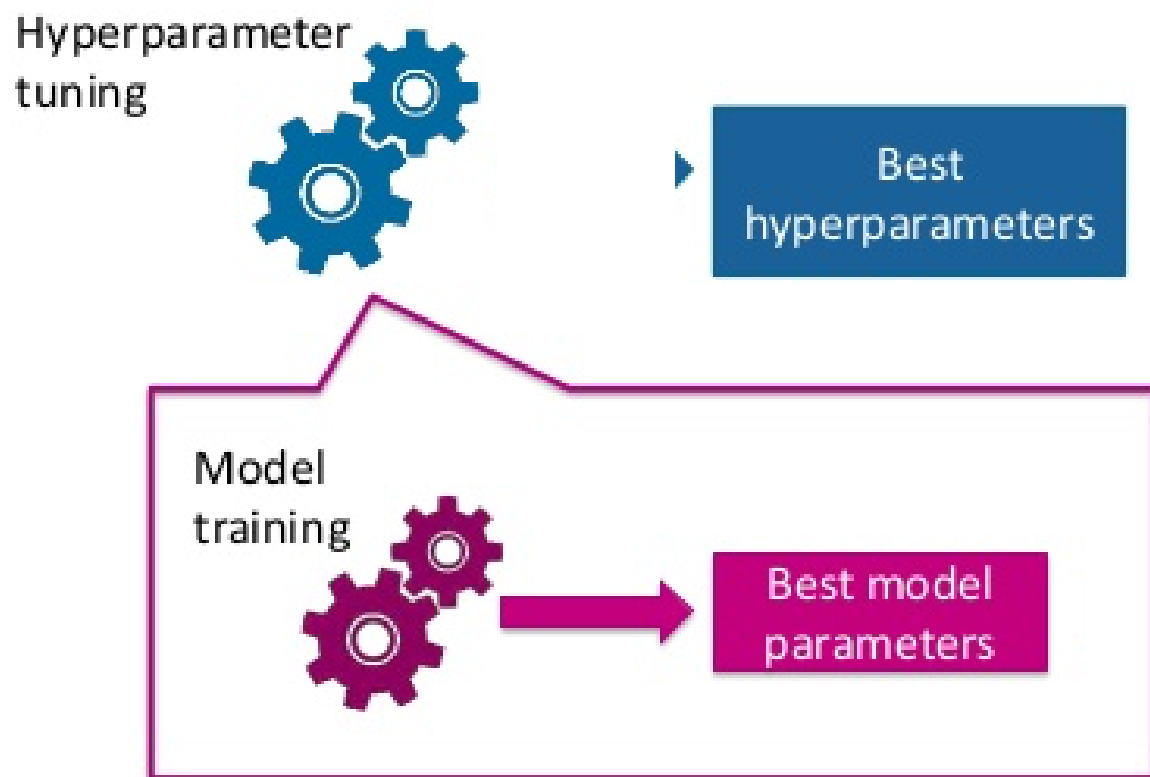
G - Gradient Descent

Optimization method to minimize error in training.

- Drives the learning process by adjusting weights.
- Central to every fine-tuning process.
- Requires proper tuning of learning rates.



H - Hyperparameters



Source

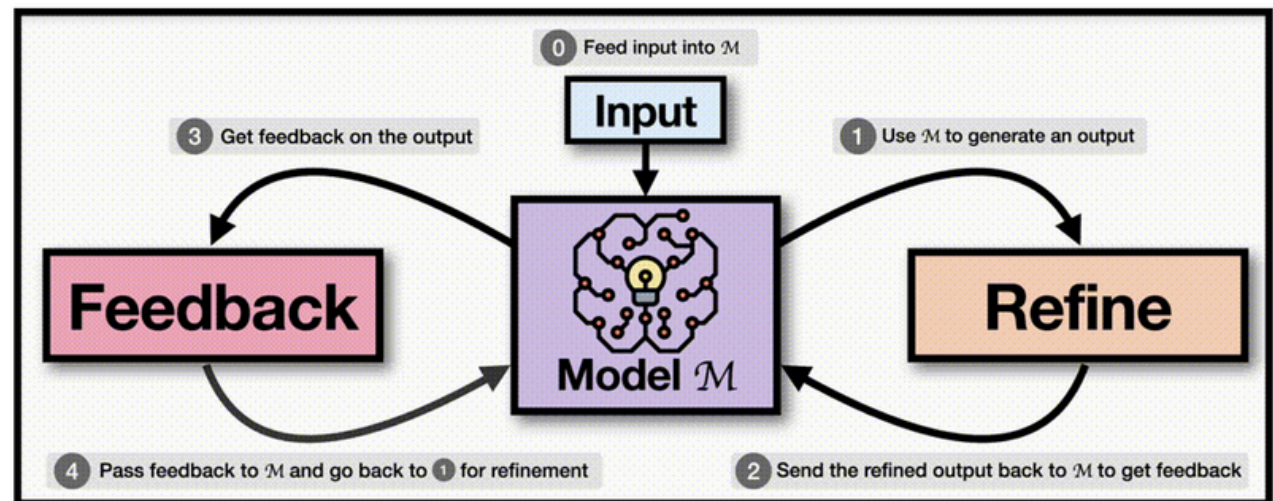
Settings like batch size, learning rate, and epochs.

- Fine-tuning requires precise hyperparameter optimization.
- Impacts model accuracy and speed.
- Experimentation helps achieve optimal results.

I - Iterative Training

Refining models through repeated training cycles.

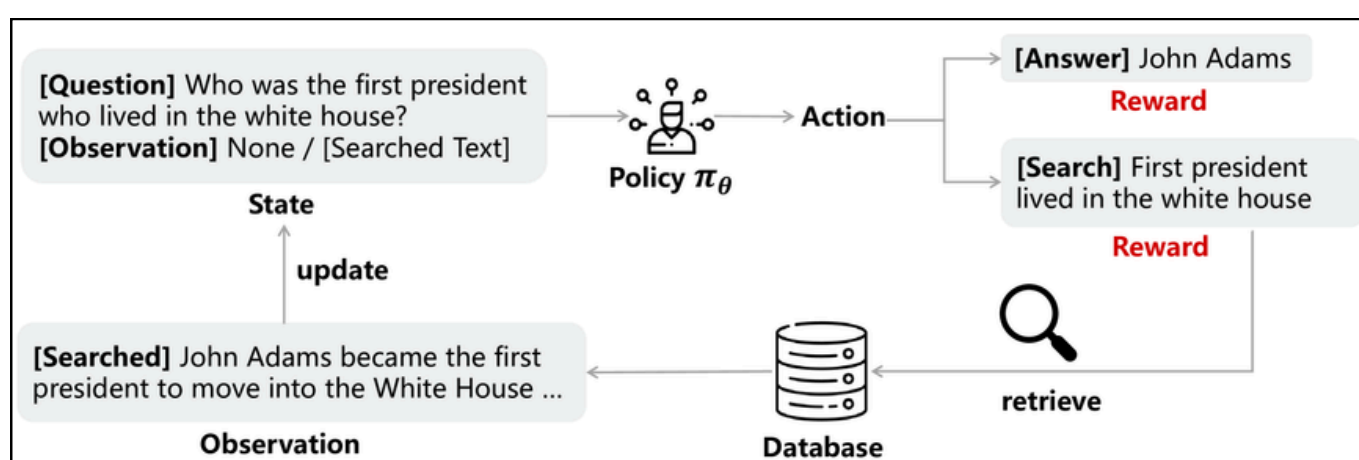
- Enhances performance step by step.
- Key for achieving state-of-the-art results.
- Reduces overfitting by monitoring progress.



Source

J - Joint Learning

Trains retrieval and generation components simultaneously.



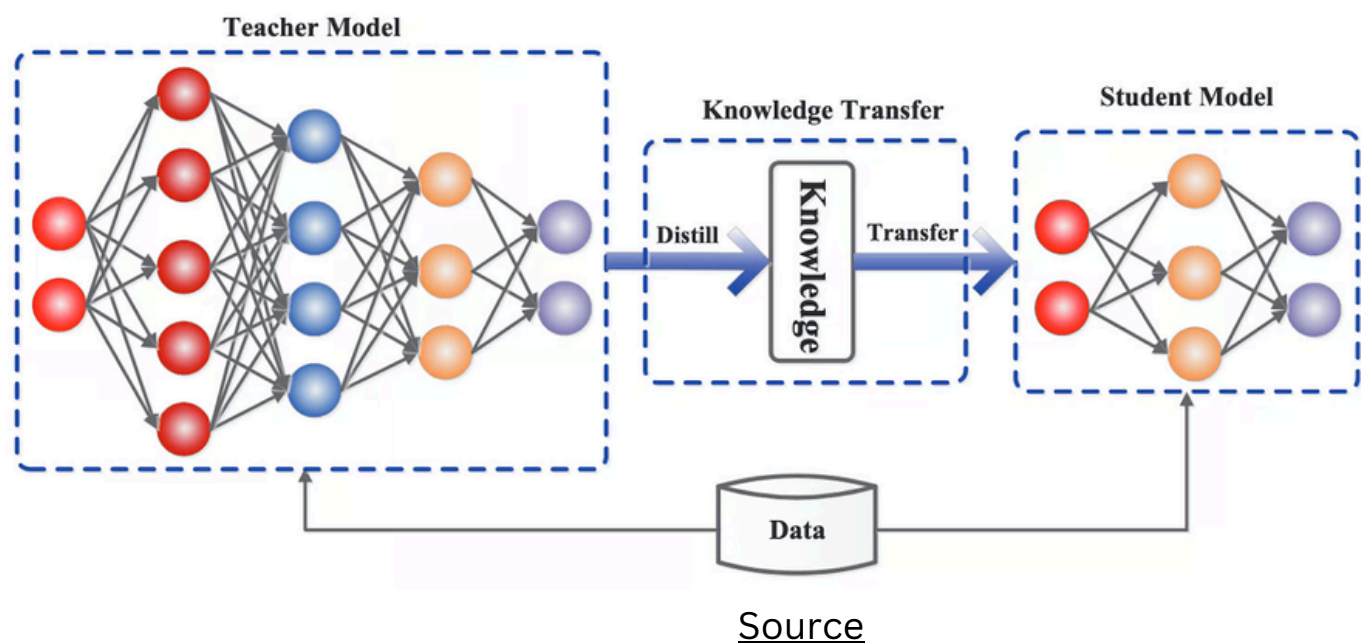
Credit:arxiv.org

- Ensures better synergy between retrieved data and generated outputs.
- Reduces the need for separate fine-tuning.
- Often leads to improved overall system performance.

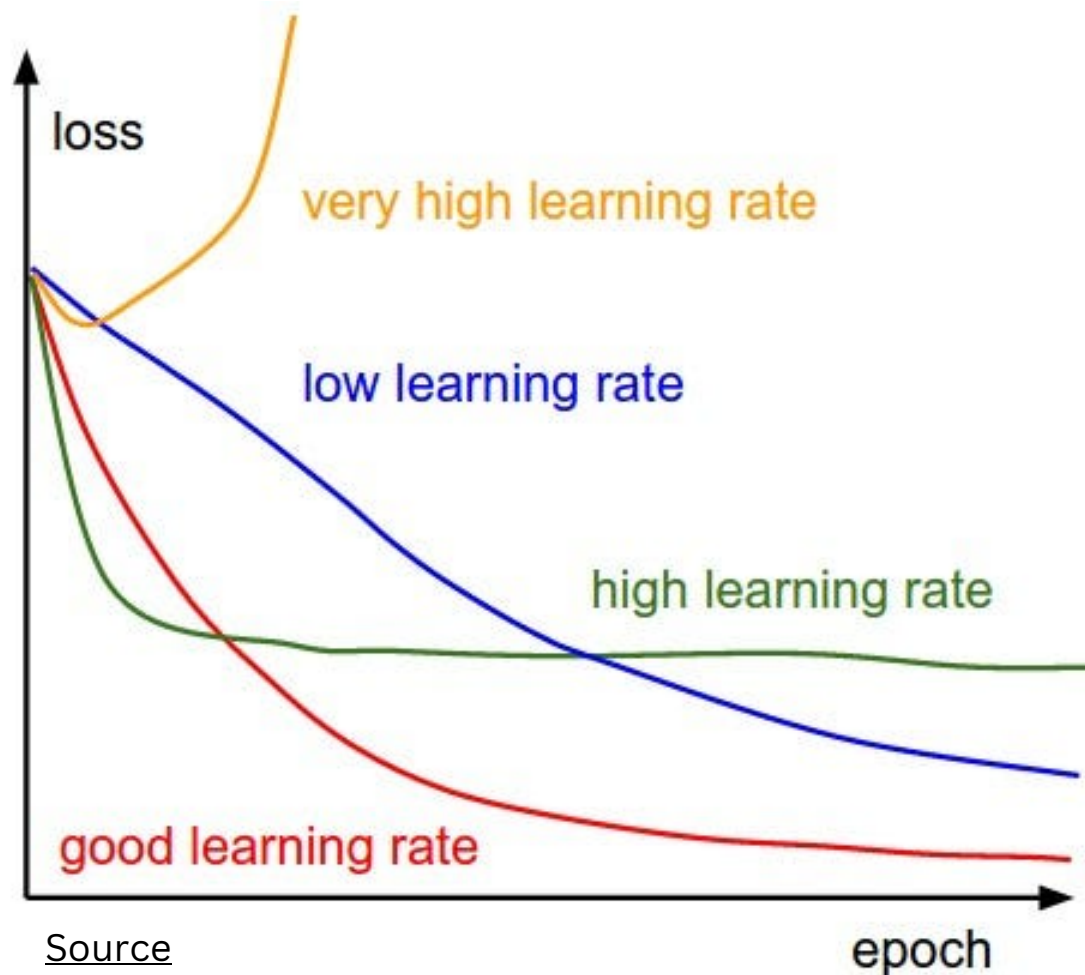
K - Knowledge Distillation

Transferring knowledge from large models to smaller ones.

- Makes models more efficient for deployment.
- Retains essential capabilities with fewer resources.
- Common in low-resource environments.



L - Learning Rate



The speed at which the model learns.

- Critical to balance between slow and fast learning.
- Improper tuning can lead to overfitting or underperformance.
- Often adjusted dynamically during training.

M - Model Weights

Parameters learned during training.

- Define how the model processes input.
- Fine-tuning adjusts weights for specific tasks.
- Pretrained weights act as a foundation.

Finetuned Weights

Weight Update

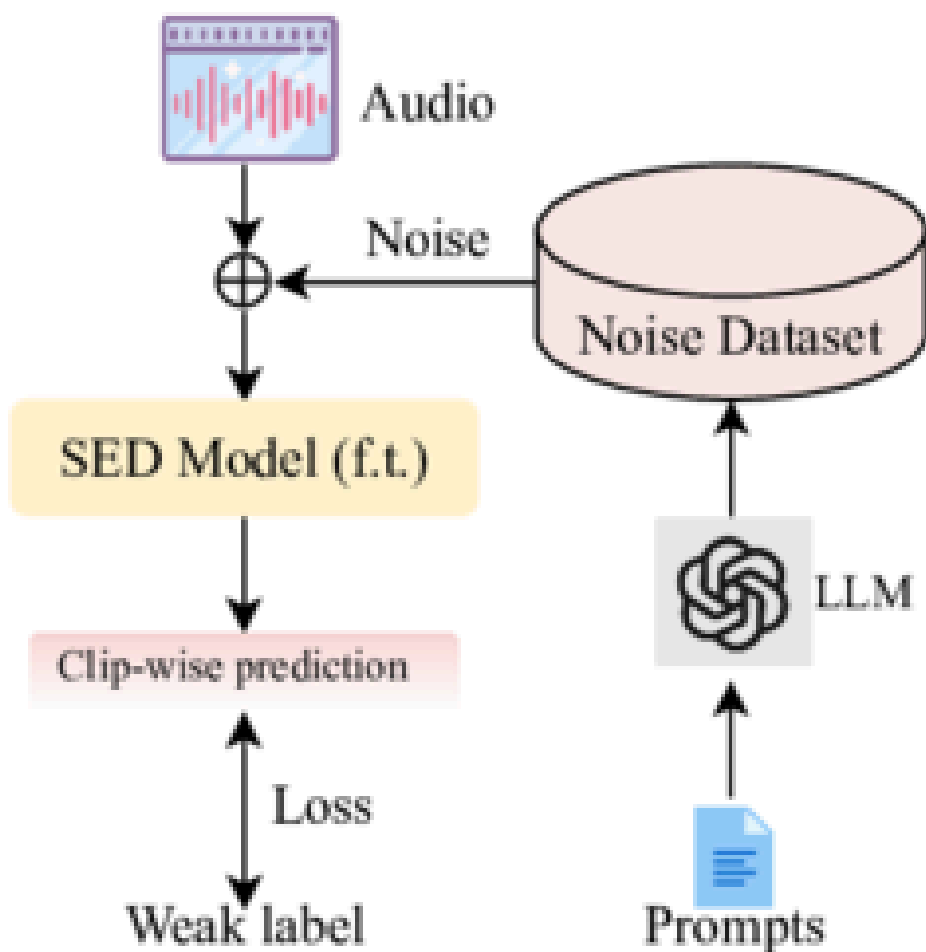
$$\overbrace{W_{ft}}^{\text{Finetuned Weights}} = \underbrace{W_{pt}}_{\text{Pretrained Weights}} + \overbrace{\Delta W}^{\text{Weight Update}}$$

Pretrained Weights

Source

N - Noise Handling

Addressing noisy or low-quality data in fine-tuning.

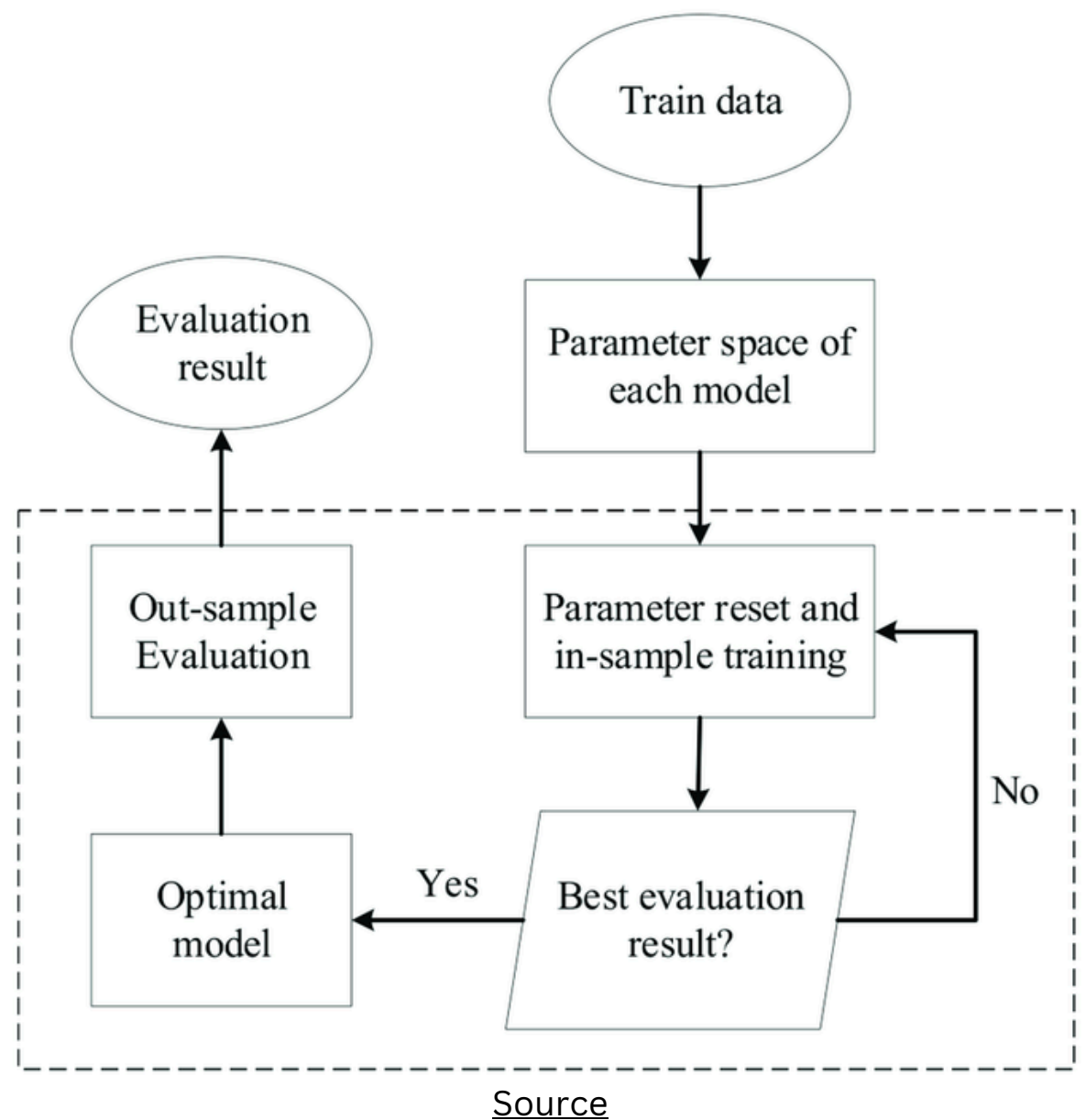


Source

O - Optimization

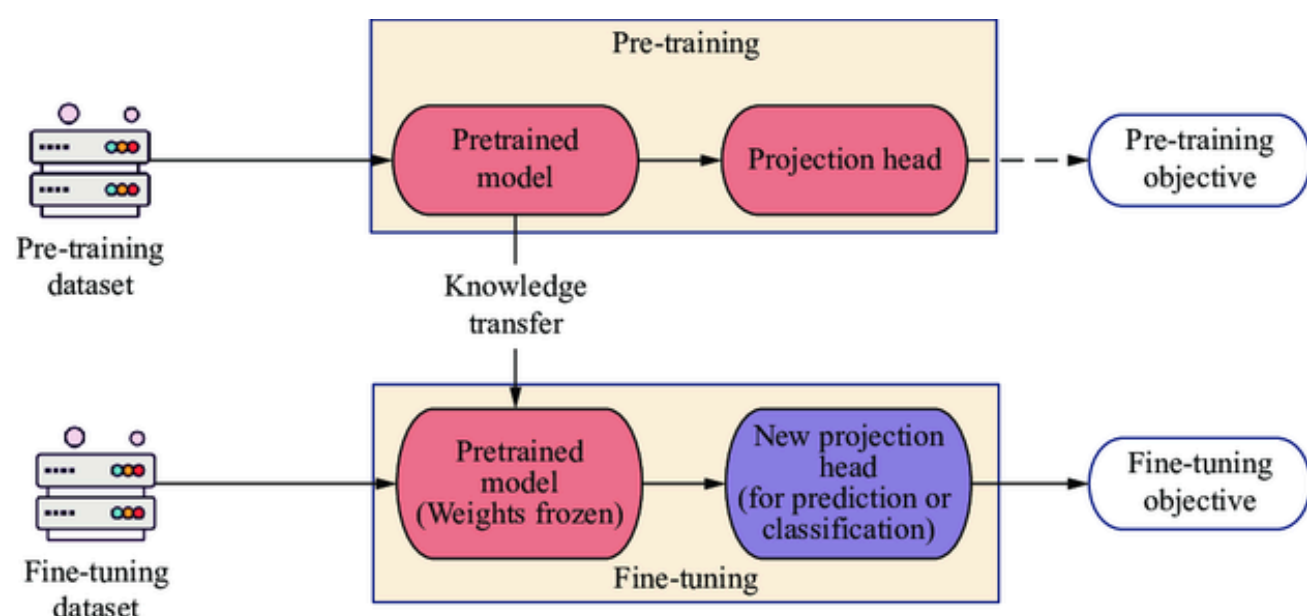
The process of improving model performance.

- Involves methods like Adam or SGD.
- Essential for faster convergence in fine-tuning.
- Balances trade-offs between accuracy and efficiency.



P - Pretraining

Training on large datasets before fine-tuning.



Source

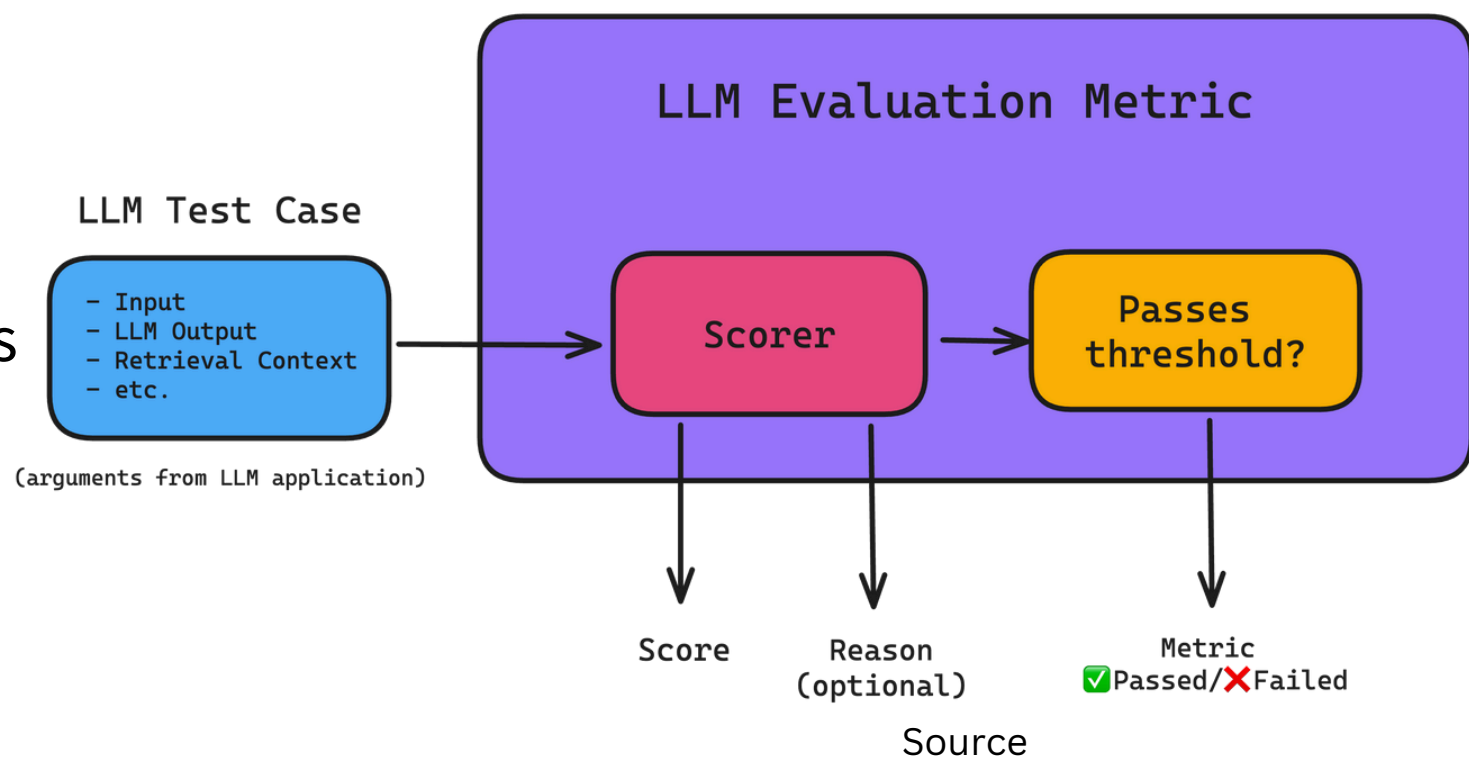
- Provides general knowledge to the model.
- Reduces data requirements for fine-tuning.
- Speeds up task-specific adaptation.

*self promotion ahead

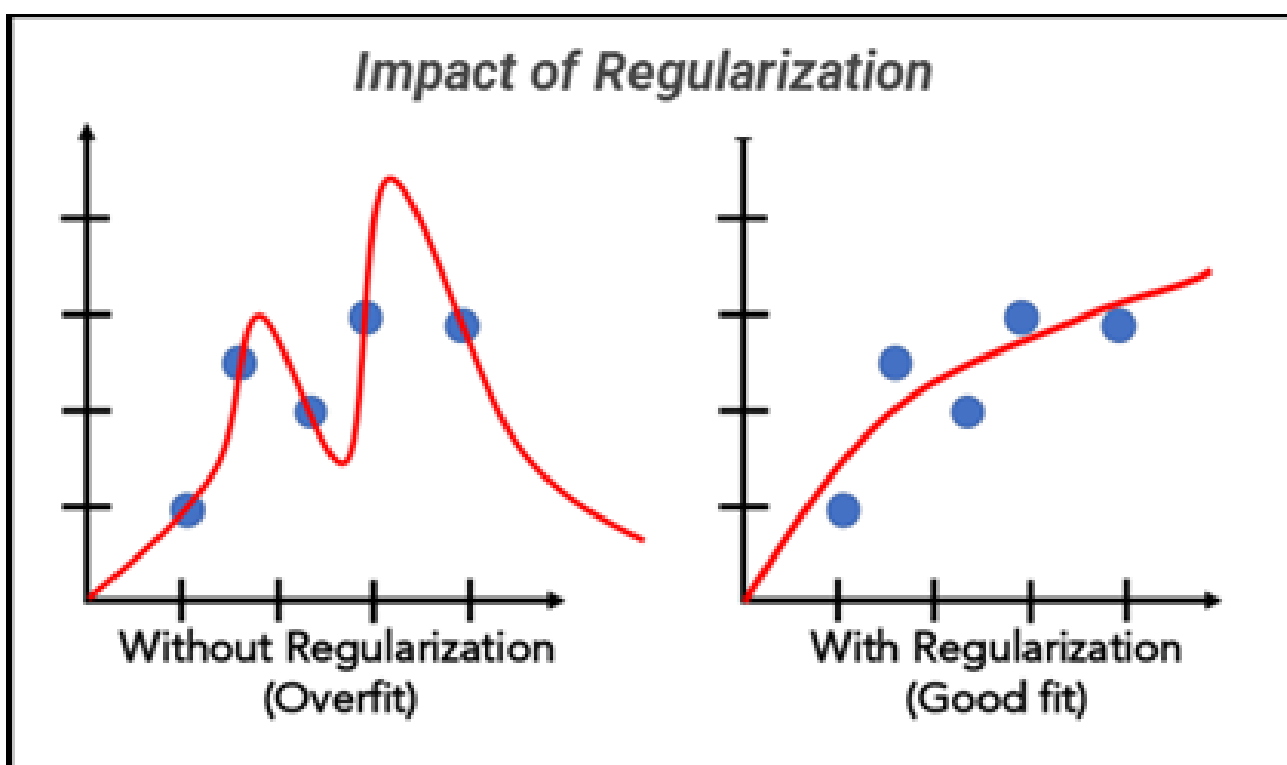
Q - Quality Evaluation

Measuring performance with metrics like BLEU, ROUGE, or accuracy.

- Validates the effectiveness of fine-tuning.
- Guides iterative improvements.
- Helps identify issues like overfitting.



R - Regularization



Source

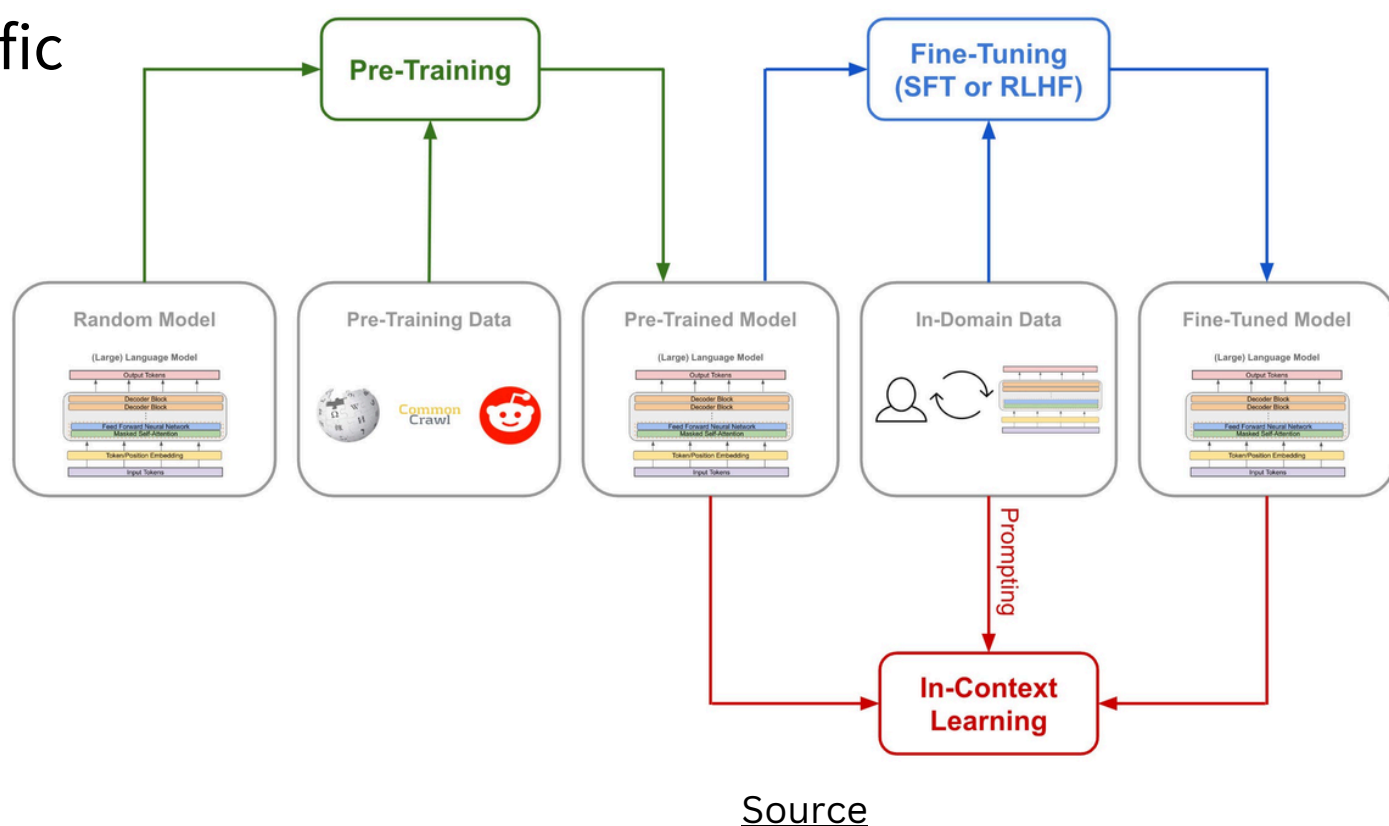
Techniques to prevent overfitting.

- Includes dropout, weight decay, or early stopping.
- Improves generalization on unseen data.
- Key for robust fine-tuned models.

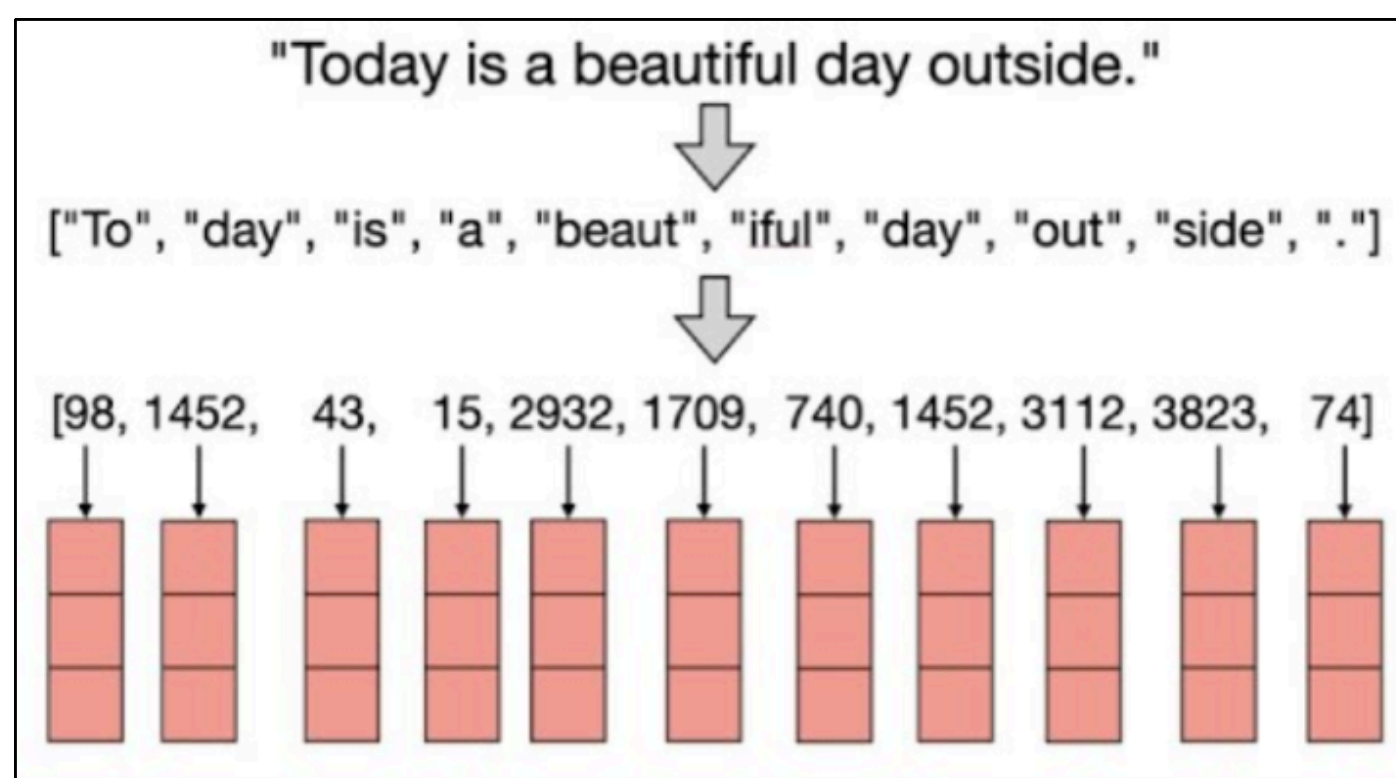
S - Supervised Fine-Tuning

Using labeled data to teach specific tasks.

- Boosts performance in well-defined use cases.
- Requires high-quality annotations.
- Common in domain-specific applications.



T - Tokenization



Splitting text into smaller units (tokens).

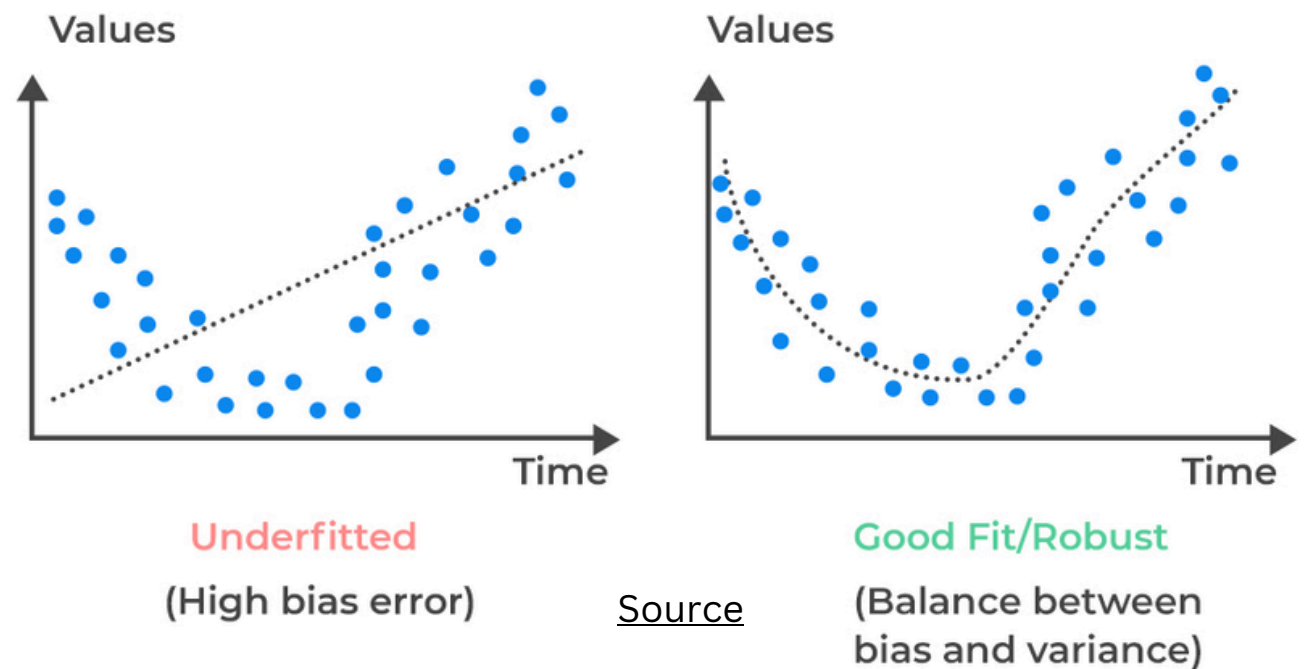
- Prepares data for model consumption.
- Handles variations like punctuation and casing.
- Fundamental for both training and inference.

Credit:Youtube.com

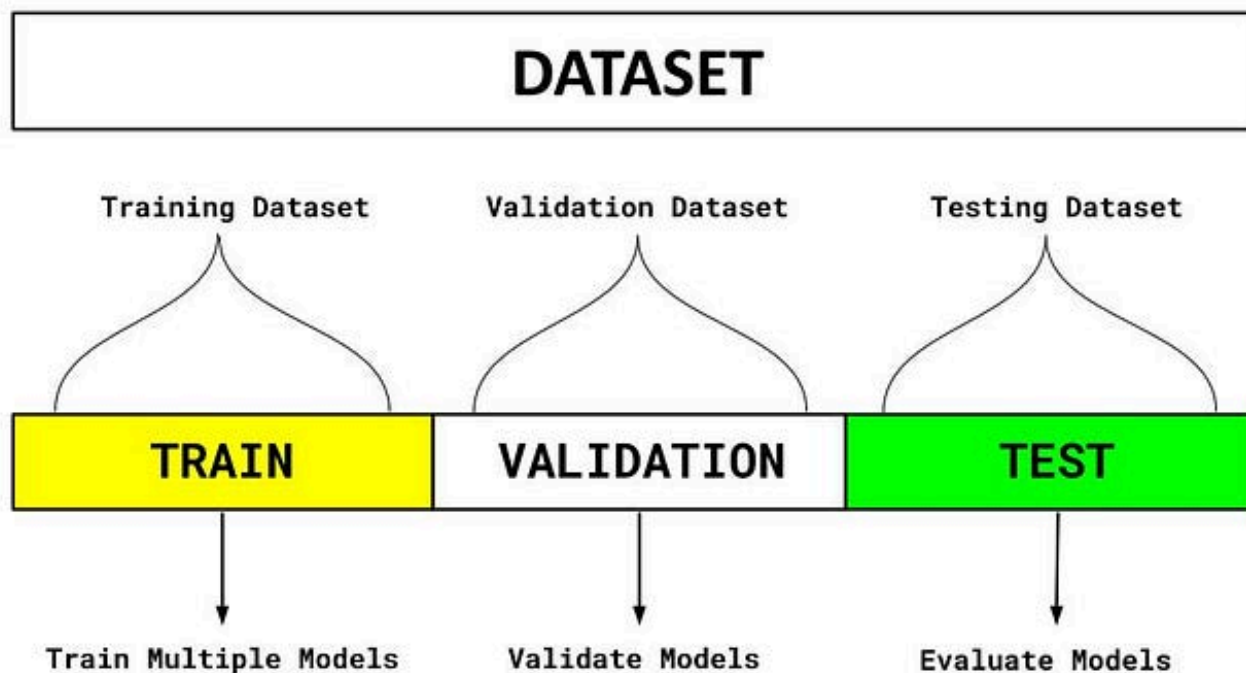
U - Underfitting

A model failing to capture data patterns.

- Often due to insufficient training.
- Addressed by increasing complexity or data size.
- Opposite of overfitting.



V - Validation Set



[Source](#)

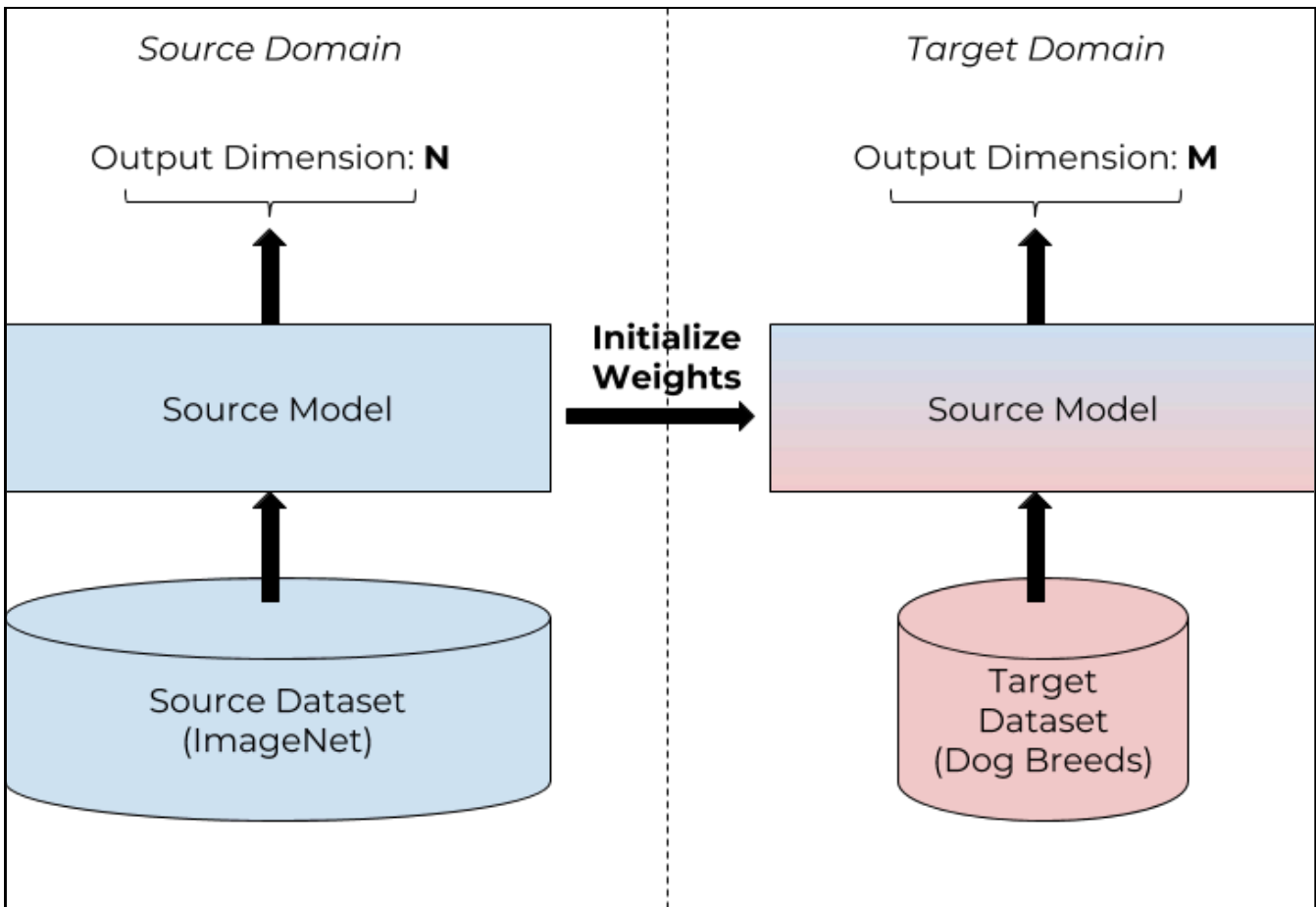
Dataset used to monitor training performance.

- Helps prevent overfitting during fine-tuning.
- Guides decisions on hyperparameter tuning.
- Ensures the model generalizes well.

W - Warm-Start

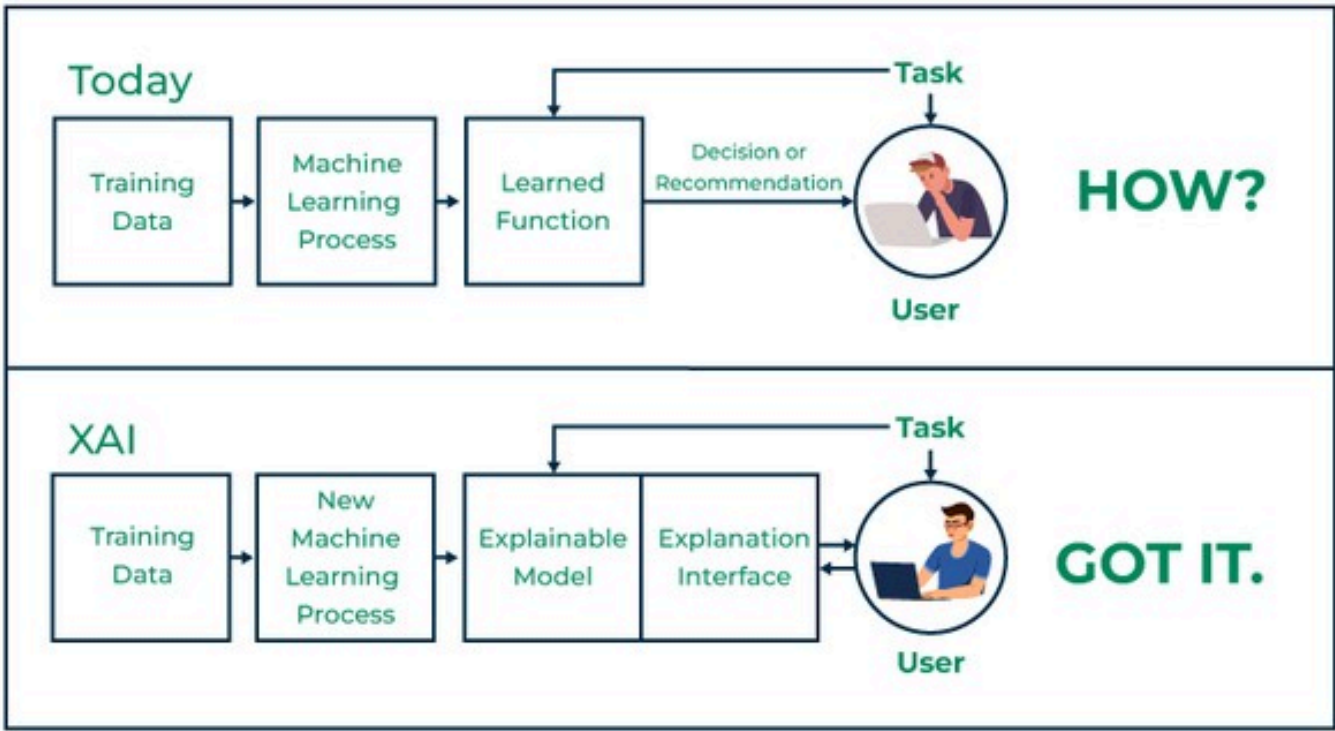
Initializes retrieval systems with pre-trained embeddings or models.

- Speeds up convergence and improves early-stage performance.
- Reduces training time for new tasks.
- Common in transfer learning scenarios.



Credit:determined.ai

X - Explainability



Credit:Medium.com

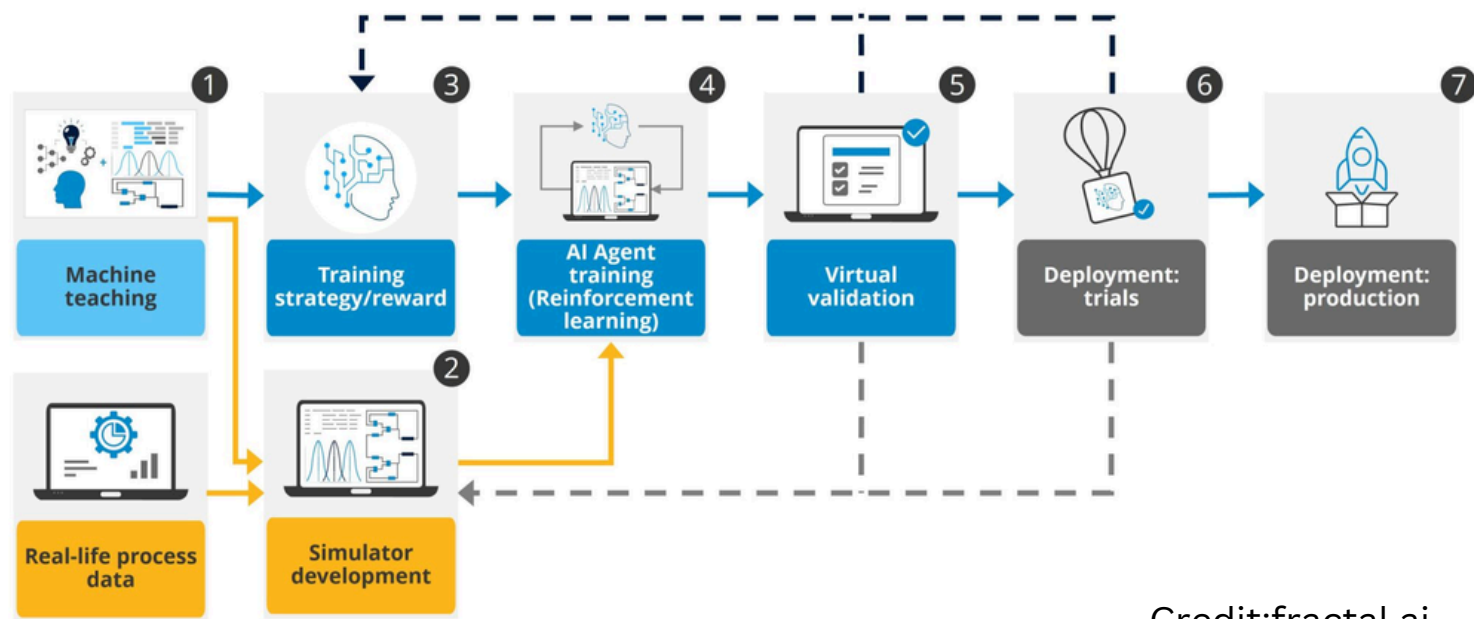
Understanding model predictions.

- Essential for building trust in AI systems.
- Identifies biases or issues in fine-tuned outputs.
- Key for high-stakes applications like healthcare.

Y - Yield Optimization

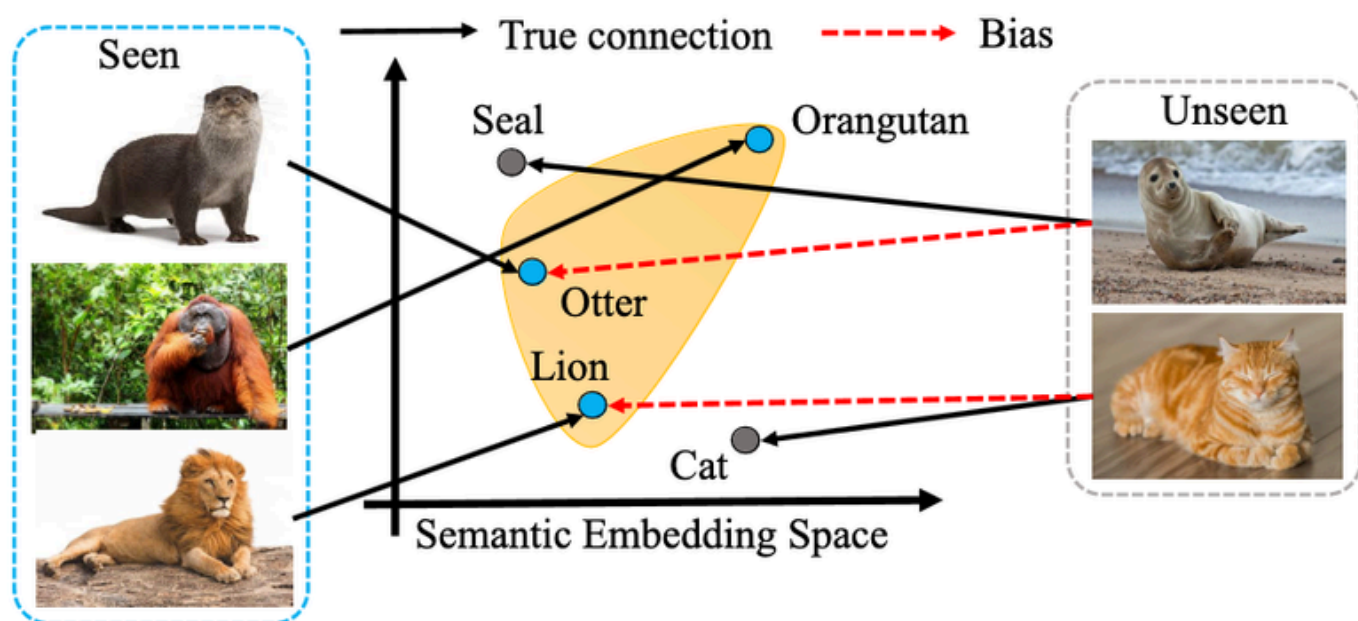
Maximizing output relevance and efficiency.

- Improves response quality for fine-tuned models.
- Involves iterative adjustments and monitoring.
- Enhances user satisfaction in real-world use.



Credit:fractal.ai

Z - Zero-shot learning



Source

Performing tasks without task-specific fine-tuning.

- Leverages general knowledge from pretraining.
- Useful for quick adaptation to new domains.
- A hallmark of advanced LLMs.